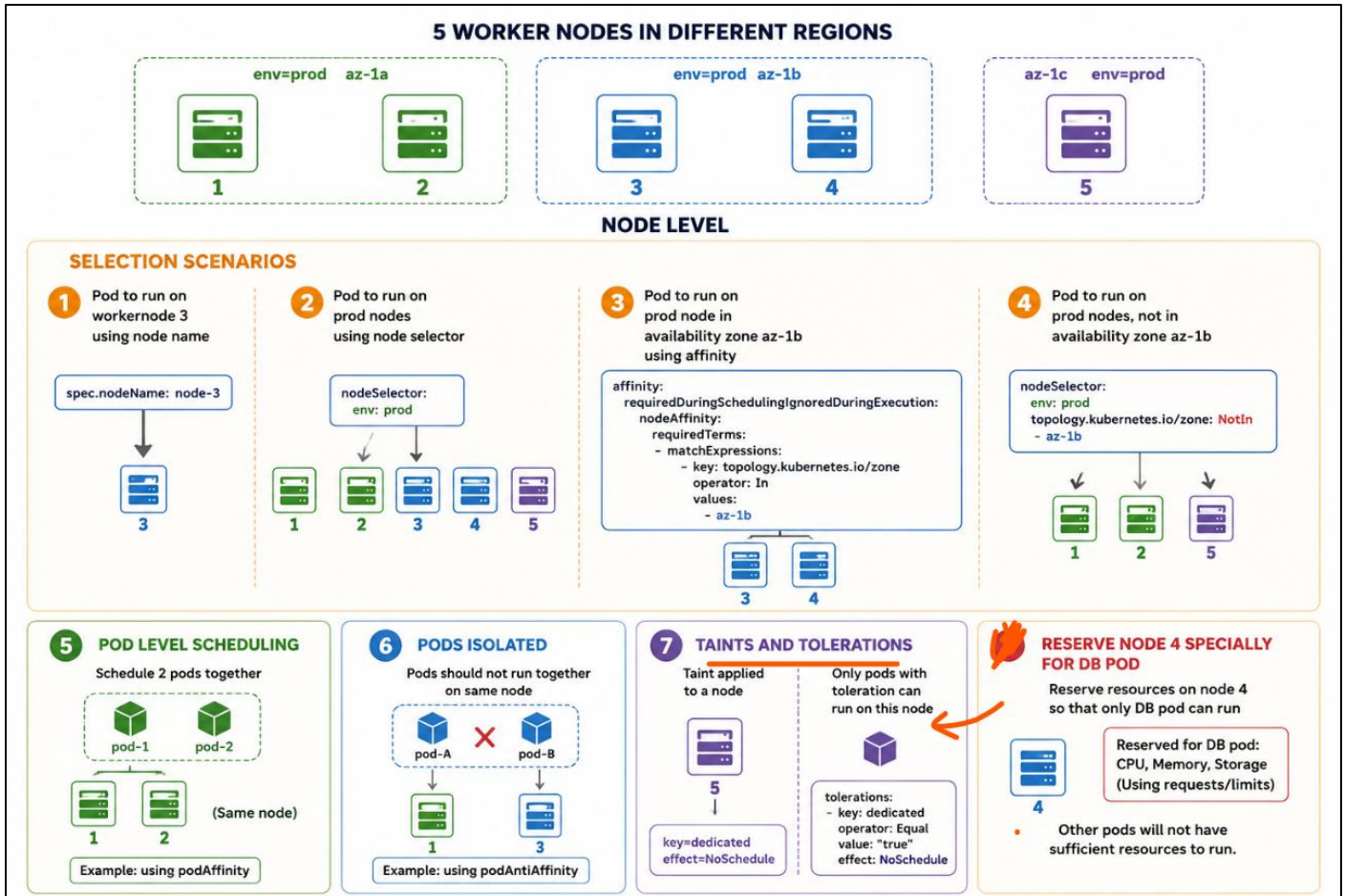


K8s Advance Scheduling



```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
$ eksctl create cluster --name clusterscheduling --nodegroup-name b17dng --node-type m7i-flex.large --nodes 4 --managed
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
$ eksctl get cluster
NAME                REGION    EKSCTL CREATED
clusterscheduling   us-east-1 True
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-19-161.ec2.internal      Ready    <none>    6m29s  v1.34.9-eks-93b80c6
ip-192-168-23-145.ec2.internal      Ready    <none>    6m29s  v1.34.9-eks-93b80c6
ip-192-168-39-27.ec2.internal       Ready    <none>    6m29s  v1.34.9-eks-93b80c6
ip-192-168-42-242.ec2.internal      Ready    <none>    6m30s  v1.34.9-eks-93b80c6
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
$ kubectl get nodes -o wide
```

NAME	STATUS	ROLES	AGE	VERSION
INTERNAL-IP	EXTERNAL-IP	OS-IMAGE		
KERNEL-VERSION	CONTAINER-RUNTIME			
ip-192-168-19-161.ec2.internal	Ready	<none>	6m51s	v1.34.9-eks-
93b80c6 192.168.19.161 34.229.149.6	Amazon Linux			
2023.12.20260611 6.12.90-120.164.amzn2023.x86_64				
containerd://2.2.4+unknown				
ip-192-168-23-145.ec2.internal	Ready	<none>	6m51s	v1.34.9-eks-
93b80c6 192.168.23.145 18.205.245.100	Amazon Linux			
2023.12.20260611 6.12.90-120.164.amzn2023.x86_64				
containerd://2.2.4+unknown				
ip-192-168-39-27.ec2.internal	Ready	<none>	6m51s	v1.34.9-eks-
93b80c6 192.168.39.27 54.242.8.96	Amazon Linux			
2023.12.20260611 6.12.90-120.164.amzn2023.x86_64				
containerd://2.2.4+unknown				
ip-192-168-42-242.ec2.internal	Ready	<none>	6m52s	v1.34.9-eks-
93b80c6 192.168.42.242 34.204.70.32	Amazon Linux			
2023.12.20260611 6.12.90-120.164.amzn2023.x86_64				
containerd://2.2.4+unknown				

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)

\$ kubectl get nodes --show-labels

NAME	STATUS	ROLES	AGE	VERSION
LABELS				
ip-192-168-19-161.ec2.internal	Ready	<none>	8m35s	v1.34.9-eks-
93b80c6 alpha.eksctl.io/cluster-				
name=clusterscheduling,alpha.eksctl.io/nodegroup-				
name=b17dng,beta.kubernetes.io/arch=amd64,beta.kubernetes.io/instance-				
type=m7i-				
flex.large,beta.kubernetes.io/os=linux,eks.amazonaws.com/capacityType=ON				
_DEMAND,eks.amazonaws.com/nodegroup-image=ami-				
0b7b393f2fbf6f220,eks.amazonaws.com/nodegroup=b17dng,eks.amazonaws.com/s				
ourceLaunchTemplateId=lt-				
058bda0b95b910f52,eks.amazonaws.com/sourceLaunchTemplateVersion=1,failur				
e-domain.beta.kubernetes.io/region=us-east-1,failure-				
domain.beta.kubernetes.io/zone=us-east-1c,k8s.io/cloud-provider-				
aws=fd98fbb4fd5de6f3d0ec160bd5a77ada,kubernetes.io/arch=amd64,kubernetes				
.io/hostname=ip-192-168-19-				
161.ec2.internal,kubernetes.io/os=linux,node.kubernetes.io/instance-				
type=m7i-flex.large,topology.k8s.aws/zone-id=use1-				
az4,topology.kubernetes.io/region=us-east-				
1,topology.kubernetes.io/zone=us-east-1c				
ip-192-168-23-145.ec2.internal	Ready	<none>	8m35s	v1.34.9-eks-
93b80c6 alpha.eksctl.io/cluster-				
name=clusterscheduling,alpha.eksctl.io/nodegroup-				
name=b17dng,beta.kubernetes.io/arch=amd64,beta.kubernetes.io/instance-				
type=m7i-				
flex.large,beta.kubernetes.io/os=linux,eks.amazonaws.com/capacityType=ON				
_DEMAND,eks.amazonaws.com/nodegroup-image=ami-				
0b7b393f2fbf6f220,eks.amazonaws.com/nodegroup=b17dng,eks.amazonaws.com/s				
ourceLaunchTemplateId=lt-				
058bda0b95b910f52,eks.amazonaws.com/sourceLaunchTemplateVersion=1,failur				
e-domain.beta.kubernetes.io/region=us-east-1,failure-				
domain.beta.kubernetes.io/zone=us-east-1c,k8s.io/cloud-provider-				
aws=fd98fbb4fd5de6f3d0ec160bd5a77ada,kubernetes.io/arch=amd64,kubernetes				
.io/hostname=ip-192-168-23-				
145.ec2.internal,kubernetes.io/os=linux,node.kubernetes.io/instance-				
type=m7i-flex.large,topology.k8s.aws/zone-id=use1-				
az4,topology.kubernetes.io/region=us-east-				
1,topology.kubernetes.io/zone=us-east-1c				
ip-192-168-39-27.ec2.internal	Ready	<none>	8m35s	v1.34.9-eks-
93b80c6 alpha.eksctl.io/cluster-				
name=clusterscheduling,alpha.eksctl.io/nodegroup-				

```

name=b17dng,beta.kubernetes.io/arch=amd64,beta.kubernetes.io/instance-
type=m7i-
flex.large,beta.kubernetes.io/os=linux,eks.amazonaws.com/capacityType=ON
_DEMAND,eks.amazonaws.com/nodegroup-image=ami-
0b7b393f2fbf6f220,eks.amazonaws.com/nodegroup=b17dng,eks.amazonaws.com/s
ourceLaunchTemplateId=lt-
058bda0b95b910f52,eks.amazonaws.com/sourceLaunchTemplateVersion=1,failur
e-domain.beta.kubernetes.io/region=us-east-1,failure-
domain.beta.kubernetes.io/zone=us-east-1d,k8s.io/cloud-provider-
aws=fd98fbb4fd5de6f3d0ec160bd5a77ada,kubernetes.io/arch=amd64,kubernetes
.io/hostname=ip-192-168-39-
27.ec2.internal,kubernetes.io/os=linux,node.kubernetes.io/instance-
type=m7i-flex.large,topology.k8s.aws/zone-id=use1-
az6,topology.kubernetes.io/region=us-east-
1,topology.kubernetes.io/zone=us-east-1d
ip-192-168-42-242.ec2.internal    Ready    <none>    8m36s    v1.34.9-eks-
93b80c6    alpha.eksctl.io/cluster-
name=clusterscheduling,alpha.eksctl.io/nodegroup-
name=b17dng,beta.kubernetes.io/arch=amd64,beta.kubernetes.io/instance-
type=m7i-
flex.large,beta.kubernetes.io/os=linux,eks.amazonaws.com/capacityType=ON
_DEMAND,eks.amazonaws.com/nodegroup-image=ami-
0b7b393f2fbf6f220,eks.amazonaws.com/nodegroup=b17dng,eks.amazonaws.com/s
ourceLaunchTemplateId=lt-
058bda0b95b910f52,eks.amazonaws.com/sourceLaunchTemplateVersion=1,failur
e-domain.beta.kubernetes.io/region=us-east-1,failure-
domain.beta.kubernetes.io/zone=us-east-1d,k8s.io/cloud-provider-
aws=fd98fbb4fd5de6f3d0ec160bd5a77ada,kubernetes.io/arch=amd64,kubernetes
.io/hostname=ip-192-168-42-
242.ec2.internal,kubernetes.io/os=linux,node.kubernetes.io/instance-
type=m7i-flex.large,topology.k8s.aws/zone-id=use1-
az6,topology.kubernetes.io/region=us-east-
1,topology.kubernetes.io/zone=us-east-1d

```

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice (main)

```

```

$ ll
total 20
drwxr-xr-x 1 Bhavana 197121 0 Jun 14 22:22 '11_June_And_12_June - CICD
Project'/
drwxr-xr-x 1 Bhavana 197121 0 Jun 21 13:11 15_june_And_16_June/
drwxr-xr-x 1 Bhavana 197121 0 Jun  2 13:21 1_june/
drwxr-xr-x 1 Bhavana 197121 0 Jun  2 11:39 29_May/
drwxr-xr-x 1 Bhavana 197121 0 Jun  3 19:56 2_june/
drwxr-xr-x 1 Bhavana 197121 0 Jun  4 07:48 3_June/
drwxr-xr-x 1 Bhavana 197121 0 Jun 11 16:54 5_june_And_9_June/

```

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice (main)
$ cd 15_june_And_16_June/

```

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```

```

$ ll
total 2797
-rw-r--r-- 1 Bhavana 197121    279 Jun 20 23:11 1-nodename.yaml
-rw-r--r-- 1 Bhavana 197121    205 Jun 20 23:11 2-nodeselector.yaml
-rw-r--r-- 1 Bhavana 197121    519 Jun 21 12:35 3-nodeaffinity-
hard.yaml
-rw-r--r-- 1 Bhavana 197121   1116 Jun 20 23:32 4-nodeaffinity-
soft.yaml
-rw-r--r-- 1 Bhavana 197121    192 Jun 20 23:43 5-backendpod.yaml
-rw-r--r-- 1 Bhavana 197121    545 Jun 20 23:35 6-podaffinity.yaml
-rw-r--r-- 1 Bhavana 197121    717 Jun 21 10:58 7-podAntiAffinity.yaml

```

```
-rw-r--r-- 1 Bhavana 197121      240 Jun 21 12:49 8-podtolerations.yaml
-rw-r--r-- 1 Bhavana 197121 1446196 Jun 21 13:11 Multi-scheduling.png
-rw-r--r-- 1 Bhavana 197121 1395555 Jun 21 12:59 scheduling.png
```

```
#1-nodename.yaml
# purpose: to create a pod and run that pod on a specific node
---
apiVersion: v1
kind: Pod
metadata:
  name: pod1
spec:
  nodeName: ip-192-168-19-161.ec2.internal
  containers:
  - name: pod-nginx-container
    image: nginx
    ports:
    - containerPort: 80
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl apply -f 1-nodename.yaml
pod/pod1 created
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl get po
```

NAME	READY	STATUS	RESTARTS	AGE
pod1	1/1	Running	0	7s

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE
NOMINATED	NODE	READINESS	GATES			
pod1	1/1	Running	0	15s	192.168.0.232	ip-192-168-19-161.ec2.internal

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl apply -f 1-nodename.yaml
pod/pod5 created
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE
NOMINATED	NODE	READINESS	GATES			
pod1	1/1	Running	0	2m22s	192.168.0.232	ip-192-168-19-161.ec2.internal
pod2	1/1	Running	0	62s	192.168.19.0	ip-192-168-19-161.ec2.internal
pod3	1/1	Running	0	44s	192.168.24.24	ip-192-168-19-161.ec2.internal
pod4	1/1	Running	0	25s	192.168.3.52	ip-192-168-19-161.ec2.internal
pod5	1/1	Running	0	7s	192.168.22.46	ip-192-168-19-161.ec2.internal


```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl apply -f 1-nodename.yaml  
pod/pod6 created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE
NOMINATED	NODE	READINESS	GATES			
pod1	1/1	Running	0	2m48s	192.168.0.232	ip-192-168-19-161.ec2.internal
		<none>		<none>		
pod2	1/1	Running	0	88s	192.168.19.0	ip-192-168-19-161.ec2.internal
		<none>		<none>		
pod3	1/1	Running	0	70s	192.168.24.24	ip-192-168-19-161.ec2.internal
		<none>		<none>		
pod4	1/1	Running	0	51s	192.168.3.52	ip-192-168-19-161.ec2.internal
		<none>		<none>		
pod5	1/1	Running	0	33s	192.168.22.46	ip-192-168-19-161.ec2.internal
		<none>		<none>		
pod6	1/1	Running	0	5s	192.168.6.27	ip-192-168-19-161.ec2.internal
		<none>		<none>		

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl delete po --all
```

```
pod "pod1" deleted from default namespace  
pod "pod2" deleted from default namespace  
pod "pod3" deleted from default namespace  
pod "pod4" deleted from default namespace  
pod "pod5" deleted from default namespace  
pod "pod6" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl get po
```

```
No resources found in default namespace.
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl label nodes ip-192-168-19-161.ec2.internal env=prod  
node/ip-192-168-19-161.ec2.internal labeled
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl label nodes ip-192-168-23-145.ec2.internal env=prod  
node/ip-192-168-23-145.ec2.internal labelled
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl label nodes ip-192-168-39-27.ec2.internal env=prod  
node/ip-192-168-39-27.ec2.internal labelled
```

```
#2-nodeselector.yaml
```

```
---  
apiVersion: v1  
kind: Pod  
metadata:  
  name: pod1-nodeselector  
spec:  
  nodeSelector:  
    env: prod
```

```
containers:
- name: pod-nginx
  image: nginx
  ports:
- containerPort: 80
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl apply -f 2-nodeselector.yaml
pod/pod1-nodeselector created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get po -o wide
NAME                                READY    STATUS    RESTARTS    AGE    IP
NODE                                NOMINATED NODE    READINESS GATES
pod1-nodeselector                  1/1      Running    0           15s    192.168.0.232
ip-192-168-19-161.ec2.internal    <none>    <none>
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl apply -f 2-nodeselector.yaml
pod/pod2-nodeselector created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get po -o wide
NAME                                READY    STATUS    RESTARTS    AGE    IP
NODE                                NOMINATED NODE    READINESS GATES
pod1-nodeselector                  1/1      Running    0           61s    192.168.0.232
ip-192-168-19-161.ec2.internal    <none>    <none>
pod2-nodeselector                  1/1      Running    0           7s     192.168.38.41
ip-192-168-39-27.ec2.internal    <none>    <none>
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl apply -f 2-nodeselector.yaml
pod/pod3-nodeselector created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get po -o wide
NAME                                READY    STATUS    RESTARTS    AGE    IP
NODE                                NOMINATED NODE    READINESS GATES
pod1-nodeselector                  1/1      Running    0           85s    192.168.0.232
ip-192-168-19-161.ec2.internal    <none>    <none>
pod2-nodeselector                  1/1      Running    0           31s    192.168.38.41
ip-192-168-39-27.ec2.internal    <none>    <none>
pod3-nodeselector                  1/1      Running    0           5s     192.168.3.52
ip-192-168-19-161.ec2.internal    <none>    <none>
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl delete po --all
pod "pod1-nodeselector" deleted from default namespace
pod "pod2-nodeselector" deleted from default namespace
pod "pod3-nodeselector" deleted from default namespace
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

\$ kubectl get nodes --show-labels

NAME	STATUS	ROLES	AGE	VERSION
ip-192-168-19-161.ec2.internal	Ready	<none>	24m	v1.34.9-eks-93b80c6
alpha.eksctl.io/cluster-name=clusterscheduling, alpha.eksctl.io/nodegroup-name=b17dng, beta.kubernetes.io/arch=amd64, beta.kubernetes.io/instance-type=m7i-flex.large, beta.kubernetes.io/os=linux, eks.amazonaws.com/capacityType=ON_DEMAND, eks.amazonaws.com/nodegroup-image=ami-0b7b393f2fbf6f220, eks.amazonaws.com/nodegroup=b17dng, eks.amazonaws.com/sourceLaunchTemplateId=lt-058bda0b95b910f52, eks.amazonaws.com/sourceLaunchTemplateVersion=1, env=prod, failure-domain.beta.kubernetes.io/region=us-east-1, failure-domain.beta.kubernetes.io/zone=us-east-1c, k8s.io/cloud-provider-aws=fd98fbb4fd5de6f3d0ec160bd5a77ada, kubernetes.io/arch=amd64, kubernetes.io/hostname=ip-192-168-19-161.ec2.internal, kubernetes.io/os=linux, node.kubernetes.io/instance-type=m7i-flex.large, topology.k8s.aws/zone-id=use1-az4, topology.kubernetes.io/region=us-east-1, topology.kubernetes.io/zone=us-east-1c				

ip-192-168-23-145.ec2.internal	Ready	<none>	24m	v1.34.9-eks-93b80c6
alpha.eksctl.io/cluster-name=clusterscheduling, alpha.eksctl.io/nodegroup-name=b17dng, beta.kubernetes.io/arch=amd64, beta.kubernetes.io/instance-type=m7i-flex.large, beta.kubernetes.io/os=linux, eks.amazonaws.com/capacityType=ON_DEMAND, eks.amazonaws.com/nodegroup-image=ami-0b7b393f2fbf6f220, eks.amazonaws.com/nodegroup=b17dng, eks.amazonaws.com/sourceLaunchTemplateId=lt-058bda0b95b910f52, eks.amazonaws.com/sourceLaunchTemplateVersion=1, env=prod, failure-domain.beta.kubernetes.io/region=us-east-1, failure-domain.beta.kubernetes.io/zone=us-east-1c, k8s.io/cloud-provider-aws=fd98fbb4fd5de6f3d0ec160bd5a77ada, kubernetes.io/arch=amd64, kubernetes.io/hostname=ip-192-168-23-145.ec2.internal, kubernetes.io/os=linux, node.kubernetes.io/instance-type=m7i-flex.large, topology.k8s.aws/zone-id=use1-az4, topology.kubernetes.io/region=us-east-1, topology.kubernetes.io/zone=us-east-1c				

ip-192-168-39-27.ec2.internal	Ready	<none>	24m	v1.34.9-eks-93b80c6
alpha.eksctl.io/cluster-name=clusterscheduling, alpha.eksctl.io/nodegroup-name=b17dng, beta.kubernetes.io/arch=amd64, beta.kubernetes.io/instance-type=m7i-flex.large, beta.kubernetes.io/os=linux, eks.amazonaws.com/capacityType=ON_DEMAND, eks.amazonaws.com/nodegroup-image=ami-0b7b393f2fbf6f220, eks.amazonaws.com/nodegroup=b17dng, eks.amazonaws.com/sourceLaunchTemplateId=lt-058bda0b95b910f52, eks.amazonaws.com/sourceLaunchTemplateVersion=1, env=prod, failure-domain.beta.kubernetes.io/region=us-east-1, failure-domain.beta.kubernetes.io/zone=us-east-1d, k8s.io/cloud-provider-aws=fd98fbb4fd5de6f3d0ec160bd5a77ada, kubernetes.io/arch=amd64, kubernetes.io/hostname=ip-192-168-39-27.ec2.internal, kubernetes.io/os=linux, node.kubernetes.io/instance-type=m7i-flex.large, topology.k8s.aws/zone-id=use1-az6, topology.kubernetes.io/region=us-east-1, topology.kubernetes.io/zone=us-east-1d				

ip-192-168-42-242.ec2.internal	Ready	<none>	24m	v1.34.9-eks-93b80c6
alpha.eksctl.io/cluster-				

```

name=clusterscheduling,alpha.eksctl.io/nodegroup-
name=b17dng,beta.kubernetes.io/arch=amd64,beta.kubernetes.io/instance-
type=m7i-
flex.large,beta.kubernetes.io/os=linux,eks.amazonaws.com/capacityType=ON
_DEMAND,eks.amazonaws.com/nodegroup-image=ami-
0b7b393f2fbf6f220,eks.amazonaws.com/nodegroup=b17dng,eks.amazonaws.com/s
ourceLaunchTemplateId=lt-
058bda0b95b910f52,eks.amazonaws.com/sourceLaunchTemplateVersion=1,failur
e-domain.beta.kubernetes.io/region=us-east-1,failure-
domain.beta.kubernetes.io/zone=us-east-1d,k8s.io/cloud-provider-
aws=fd98fbb4fd5de6f3d0ec160bd5a77ada,kubernetes.io/arch=amd64,kubernetes
.io/hostname=ip-192-168-42-
242.ec2.internal,kubernetes.io/os=linux,node.kubernetes.io/instance-
type=m7i-flex.large,topology.k8s.aws/zone-id=use1-
az6,topology.kubernetes.io/region=us-east-
1,topology.kubernetes.io/zone=us-east-1d

```

```

# 3-nodeaffinity.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: pod1-required-nodeaffinity
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: env
                operator: In
                values:
                  - prod
              - key: topology.kubernetes.io/zone
                operator: In
                values:
                  - us-east-1c
  containers:
    - name: nginx
      image: nginx

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl apply -f 3-nodeaffinity-hard.yaml
```

pod/pod1-required-nodeaffinity created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP
NODE	NOMINATED	NODE	READINESS	GATES	
pod1-required-nodeaffinity	1/1	Running	0	11s	
192.168.24.24	ip-192-168-19-161.ec2.internal	<none>	<none>		

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

\$ kubectl delete po --all

pod "pod1-required-nodeaffinity" deleted from default namespace

```
# 4-nodeaffinity-soft.yaml
# purpose: to create a pod with soft node affinity rules
---
apiVersion: v1
kind: Pod
metadata:
  name: pod1-required-and-preferred-nodeaffinity
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: env
                operator: In
                values:
                  - prod

        preferredDuringSchedulingIgnoredDuringExecution:
          - weight: 1
            preference:
              matchExpressions:
                - key: topology.kubernetes.io/zone
                  operator: In
                  values:
                    - us-east-1c

          - weight: 5
            preference:
              matchExpressions:
                - key: topology.kubernetes.io/zone
                  operator: In
                  values:
                    - us-east-1a

          - weight: 2
            preference:
              matchExpressions:
                - key: topology.kubernetes.io/zone
                  operator: In
                  values:
                    - us-east-1d

  containers:
    - name: pod-nginx
      image: nginx
      ports:
```

```
- containerPort: 80
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl apply -f 4-nodeaffinity-soft.yaml
pod/pod1-required-and-preferred-nodeaffinity created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS
AGE	IP	NODE	NOMINATED
READINESS GATES			
pod1-required-and-preferred-nodeaffinity	1/1	Running	0
34s	192.168.37.202	ip-192-168-39-27.ec2.internal	<none>

```
<none>
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl apply -f 4-nodeaffinity-soft.yaml
pod/pod2-required-and-preferred-nodeaffinity created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS
AGE	IP	NODE	NOMINATED
READINESS GATES			
pod1-required-and-preferred-nodeaffinity	1/1	Running	0
87s	192.168.37.202	ip-192-168-39-27.ec2.internal	<none>
<none>			
pod2-required-and-preferred-nodeaffinity	1/1	Running	0
9s	192.168.54.132	ip-192-168-39-27.ec2.internal	<none>

```
<none>
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl delete po --all
pod "pod1-required-and-preferred-nodeaffinity" deleted from default namespace
pod "pod2-required-and-preferred-nodeaffinity" deleted from default namespace
```

```
#5-backendpod.yaml
```

```
---
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: backend-pod
```

```
  labels:
```

```
    app: backend
```

```
spec:
```

```
  containers:
```

```
  - name: pod-nginx
```

```
    image: nginx
```

```
    ports:
```

```
    - containerPort: 80
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl apply -f 5-backendpod.yaml  
pod/backend-pod created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl get po -o wide  
NAME READY STATUS RESTARTS AGE IP NODE  
NOMINATED NODE READINESS GATES  
backend-pod 1/1 Running 0 9s 192.168.0.232 ip-192-168-19-161.ec2.internal <none>
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl get po --show-labels  
NAME READY STATUS RESTARTS AGE LABELS  
backend-pod 1/1 Running 0 51s app=backend
```

```
#6-podaffinity.yaml  
#Schedule frontend-pod on the same node where a pod with label app=backend is  
already running.  
---  
apiVersion: v1  
kind: Pod  
metadata:  
  name: frontend-pod  
spec:  
  affinity:  
    podAffinity:  
      requiredDuringSchedulingIgnoredDuringExecution:  
      - labelSelector:  
          matchExpressions:  
          - key: app  
            operator: In  
            values:  
            - backend  
        topologyKey: kubernetes.io/hostname  
  containers:  
  - name: nginx-container  
    image: nginx  
    ports:  
    - containerPort: 80
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl apply -f 6-podaffinity.yaml  
pod/frontend-pod created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
```

```
$ kubectl get po -o wide  
NAME READY STATUS RESTARTS AGE IP NODE  
NOMINATED NODE READINESS GATES
```

```
backend-pod      1/1      Running    0           72s    192.168.0.232    ip-
192-168-19-161.ec2.internal    <none>    <none>
frontend-pod     1/1      Running    0           5s     192.168.20.72    ip-
192-168-19-161.ec2.internal    <none>    <none>
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

\$ kubectl get po --show-labels

```
NAME          READY   STATUS    RESTARTS   AGE   LABELS
backend-pod   1/1     Running   0          79s   app=backend
frontend-pod  1/1     Running   0          12s   <none>
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

\$ kubectl delete po --all

pod "backend-pod" deleted from default namespace

pod "frontend-pod" deleted from default namespace

#7-podAntiAffinity.yaml

```
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-deployment

spec:
  replicas: 2

  selector:
    matchLabels:
      app: web
      env: prod

  template:
    metadata:
      labels:
        app: web
        env: prod

    spec:
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            - labelSelector:
                matchExpressions:
                  - key: env
                    operator: In
                    values:
                      - prod
              topologyKey: kubernetes.io/hostname

      containers:
        - name: nginx-container
          image: nginx
          ports:
            - containerPort: 80
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl apply -f 7-podAntiAffinity.yaml
deployment.apps/web-deployment created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get deploy
NAME                READY    UP-TO-DATE    AVAILABLE    AGE
web-deployment      2/2      2             2             2m8s
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get po
NAME                                READY    STATUS    RESTARTS    AGE
web-deployment-8666bddbf9-f9m79    1/1      Running   0            24s
web-deployment-8666bddbf9-rzzgm     1/1      Running   0            24s
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get po -o wide
NAME                                READY    STATUS    RESTARTS    AGE    IP
NODE                                NOMINATED NODE    READINESS GATES
web-deployment-8666bddbf9-f9m79    1/1      Running   0            33s    192.168.24.24
ip-192-168-19-161.ec2.internal      <none>
web-deployment-8666bddbf9-rzzgm     1/1      Running   0            33s    192.168.54.132
ip-192-168-39-27.ec2.internal      <none>
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get po --show-labels
NAME                                READY    STATUS    RESTARTS    AGE
LABELS
web-deployment-8666bddbf9-f9m79    1/1      Running   0            46s    app=web,env=prod,pod-template-hash=8666bddbf9
web-deployment-8666bddbf9-rzzgm     1/1      Running   0            46s    app=web,env=prod,pod-template-hash=8666bddbf9
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl delete deploy web-deployment
deployment.apps "web-deployment" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get po
No resources found in default namespace.
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)
$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-19-161.ec2.internal      Ready     <none>    58m    v1.34.9-eks-93b80c6
```



```
ip-192-168-23-145.ec2.internal    Ready    <none>    58m    v1.34.9-eks-93b80c6
ip-192-168-39-27.ec2.internal    Ready    <none>    58m    v1.34.9-eks-93b80c6
ip-192-168-42-242.ec2.internal    Ready    <none>    58m    v1.34.9-eks-93b80c6
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl taint nodes ip-192-168-42-242.ec2.internal
dedicated=database:NoSchedule
node/ip-192-168-42-242.ec2.internal tainted
```

```
#8-testpod.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: testpodfortaint
spec:
  containers:
  - name: pod-nginx-container
    image: nginx
    ports:
    - containerPort: 80
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl apply -f 8-testpod.yaml
pod/testpodfortaint created
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE
NOMINATED NODE	READINESS GATES					
testpodfortaint	1/1	Running	0	15s	192.168.6.27	ip-192-168-19-161.ec2.internal
		<none>			<none>	

```
#9-podtolerations.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: db-pod
spec:
  tolerations:
  - key: "dedicated"
    operator: "Equal"
    value: "database"
    effect: "NoSchedule"
  containers:
  - name: mysql-database
    image: nginx
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl apply -f 9-podtolerations.yaml
pod/db-pod created
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/15_june_And_16_June (main)

```
$ kubectl get po -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP
db-pod	1/1	Running	0	6s	192.168.37.202
ip-192-168-39-27.ec2.internal		<none>			<none>
testpodfortaint	1/1	Running	0	115s	192.168.6.27
ip-192-168-19-161.ec2.internal		<none>			<none>

